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Viologen−Bromide Dual-Redox Ionic Solid Complexes: Understanding Their Electrochemical Formation and Proton-Accompanied Redox Chemistry

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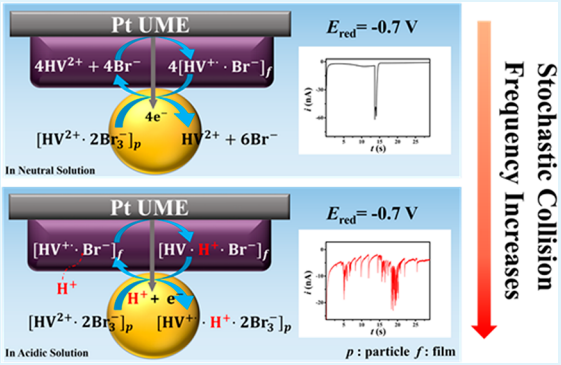
Supporting Information

ABSTRACT: The inhibition of self-discharge in a redox-enhanced electrochemical capacitor (Redox-EC) is crucial for excellent energy retention. Heptyl viologen dibromide ($HVBr_2$) was chosen as a strong candidate of a dual-redox species in Redox-EC due to its solid complexations during the charging process, at which HV^{2+} is electrochemically reduced to $HV^{+\bullet}$ and form a solid complex, $[HV^{+\bullet}{\cdot}Br^-]$, on an anode while Br^- is electro-oxidized to Br_3^- and renders $[HV^{2+}{\cdot}2Br_3^-]$ on a cathode. The solid complexes could not transfer across the separator, resulting in significant diminution of the self-discharge. In this Article, we present detailed electrochemical studies of formation of $[HV^{2+}{\cdot}2Br_3^-]$ and $[HV^{+\bullet}{\cdot}Br^-]$, their redox features, and galvanic exchange reactions between the two types of dual-redox ionic solids on a Pt ultra-microelectrode (UME) in neutral (0.33 M Na_2SO_4) and acidic (1 M H_2SO_4) solutions. Most importantly, through voltammetric and particle-impact electrochemical analyses, we found that the redox and galvanic exchange reactions of the two dual-redox ionic solid complexes involve H^+ transfer, which is the key process to limit the overall kinetics of the electrochemical reactions. We also rationalize the proton-accompanied galvanic exchange reaction based on computational simulation.



KEYWORDS: *redox-enhanced electrochemical capacitor, dual-redox ionic solid complexation, proton-accompanied redox chemistry, particle-impact electrochemistry, galvanic exchange reaction*

INTRODUCTION

The formation of redox ionic complexes (RICs) has been considered a powerful strategy for inhibition of the self-discharge reaction occurring in energy storage devices (ESDs) operated in aqueous electrolytes.[1−6] In a Zn−Br redox flow battery (RFB), this self-discharge driven by Br_2 crossover through a membrane (or separator) can be greatly suppressed by the addition of quaternary ammonium bromide (QBr) to form polybromide (QBr_{2n+1}) liquid, which is immiscible with an aqueous phase.[7−12] The electrochemically generated QBr_{2n+1} can hold Br_2 stably, resulting in a superior energy retention rate during charge/discharge cycles in the Zn−Br RFB.

In redox-enhanced electrochemical capacitors (Redox-ECs), as another example of ESDs, Stucky et al. adopted a similar strategy to the Redox-EC using Br^-/Br_2 as a half redox reaction to inhibit the self-discharge reaction.[13−15] In this system, however, they used tetrabutylammonium bromide (TBABr) as the Br_2 sequestration agent to form a stable ionic solid tribromide complex on the cathode.[13]

They also introduced dual-redox mediators of HV^{2+} and Br^- to the Redox-EC system.[14,15] During a charge process, HV^{2+} is reduced to form solid $[HV^{+\bullet}{\cdot}Br^-]$ on the anode, while Br^- is oxidized to form solid $[HV^{2+}{\cdot}2Br_3^-]$ on the cathode, resulting in significant suppression of self-discharge in the Redox-EC. However, the estimated electrochemical performances based on the suggested formation/dissolution reactions of two RICs may be oversimplified, and it is highly probable that the electrochemistry of the two RICs is more complicated and not yet known.

Our group studied the electrochemistry of QBr_{2n+1} droplets through particle-impact electrochemistry (PIE).[16−19] Furthermore, we electrochemically monitored the phase transition of ionic liquid tetraalkylammonium tribromide (TBr_3) from droplets to ionic solid particles.[20] These QBr_{2n+1} and TBr_3 species are considered single RICs, in which only the anion is electrochemically active. The present study also showed that analysis of stochastic electrochemical events associated with the collision of individual dual RIC particles on a Pt ultra-

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microelectrode (UME) is important to understand the electrochemistry of the dual RICs.

The monitoring of stochastic electrochemical responses from particles colliding on a UME, i.e., PIE, has received great attention as a new powerful method to analyze the electrochemical and physicochemical properties of individual entities, which are different from their ensembles. So far, PIE has been reported for metals,[21−36] semiconductors,[37] carbons,[38,39] insulators,[40−49] droplets,[16−19] emulsions,[50−53] micelles,[54−57] vesicles,[48,58] enzymes,[59,60] and clay particles.[61]

In PIE for insulating particles (IPs), a solution is composed of soluble redox mediators, such as $Fe(CN)^{4-}/Fe(CN)^{3-}$. When IPs are injected in the solution, they diffuse and stochastically collide on a UME, resulting in sudden current drops due to the partial blockage of active UME sites where redox reactions from mediators occur. This analytical strategy is possible because the IPs are electrochemically inactive. PIE for the characterization of IPs has been widely applied to the analyses of polystyrene particles,[40−44] silica,[42] bacteria,[46] biomacromolecules,[47] vesicles,[48] and viruses.[49]

Another type of IP is single or dual RIC particles. Unlike other IPs, which do not have their own redox properties, we can analyze the electroreduction potentials, redox kinetics, and quantity of redox ions (or molecules) of RIC particles through PIE.

To date, our group has fundamentally focused on the electrochemistry of single RICs. We now extend our scope to understanding the electrochemistry of dual RICs in this Article. We chose dual RICs composed of either $[HV^{2+}{\cdot}2Br_3^-]$ or $[HV^{+\bullet}{\cdot}Br^-]$ as models, where HV is heptyl viologen. We found that $[HV^{+\bullet}{\cdot}Br^-](s)$ formed on the Pt UME is electrochemically reduced via H^+ transfer. Also, we deciphered stochastic current spikes associated with the galvanic exchange reaction between an $[HV^{+\bullet}{\cdot}Br^-](s)$ film and an $[HV^{2+}{\cdot}2Br_3^-]$ particle and proposed a different electroreduction mechanism of $[HV^{2+}{\cdot}2Br_3^-]$ particles on a modified Pt UME with $[HV^{+\bullet}{\cdot}Br^-]$ under acidic conditions from that in a solution with neutral pH.

Based on the presented stochastic particle-impact results, even in the Redox-EC to use the dual redox-active ions, HV^{2+} and Br^- to form the two different types of dual RICs (solids) during the charging process, the self-discharge may occur if an average pore size of a separator in the Redox-EC would be larger than that of $[HV^{2+}{\cdot}2Br_3^-]$ particles generated on a cathode, which would become more significant as the pH's of both a catholyte and an anolyte decrease.

EXPERIMENTAL SECTION

Chemicals. All solutions were prepared with deionized water. Sulfuric acid (H_2SO_4, 98%) was purchased from Daejung Chemicals (Cheongwon, Korea), and 1,1′-diheptyl-4,4′-bipyridinium dibromide ($C_{24}H_{38}Br_2N_2$) was purchased from TCI Korea. Sodium sulfate (Na_2SO_4) was purchased from Sigma-Aldrich. All reagents and solvents were used without further purification.

Instruments and Measurements. A CHI-600e potentiostat (CH Instruments, Austin, TX) was used for all electrochemical measurements. Three electrodes were used in the electrochemical cells: Pt UME (radius: $a = 5\ \mu m$) as a working electrode, Ag/AgCl (1 M KCl) as a reference electrode, and Pt wire as a counter electrode. All electrodes were purchased and used from CH Instruments. The electrochemical measurements were conducted in a home-built Faradaic cage. All the aqueous solutions were deaerated with Ar for ca. 15 min, and the ionic strength, I, in the solutions was adjusted to 1 M by addition of Na_2SO_4. The scan rate was 0.02 V/s for all the voltammetric measurements. FTIR measurements were performed using a Bruker LUMOS FTIR microscope (Bruker Optics Korea) in ATR (attenuated total reflectance) mode. The sample absorbance spectra were collected from each measurement point (size = 100 μm × 100 μm) after accumulation of 32 scans with a spectral resolution of 4 cm^{-1} using an MCT (mercury cadmium telluride) detector over the range 4000−600 cm^{-1} (OPUS v7.2 software, Bruker Optik GmbH). Raman spectroscopy was carried out using a DXR Raman microscope with a 532 nm excitation laser (Thermo Scientific Co.). XPS measurements were performed with a K-alpha plus (Thermo Scientific) using Al Kα radiation (hυ = 1486.6 eV) in vacuum conditions, around 1.0×10^{-9} mbar (an analysis vacuum condition: 3.5×10^{-9} mbar) before and after sputtering with Ar^+ ions. The spectra were recorded at a pass energy of 200 eV and an energy step size of 1.0 eV for a survey scan.

Computational Simulation. The geometries of $[HV^{2+}{\cdot}2Br_3^-]$, $[HV^{+\bullet}{\cdot}Br^-]$, and $[HV^{2+}{\cdot}2Br_3^-]_2$ ion-pair model systems and their related protonated species were optimized at B3LYP[62] and M062X[63] with 6-31G(d) basis sets.[64] The total electronic energies were corrected by zero-point vibrational energies, and the free energies were computed in the gas phase and in water solvent using the continuum solvation model based on density (SMD).[65] Vibrational calculations were verified by frequency calculations to ensure that all geometries were true minima. To rationalize the formation of $[HV^{+\bullet}{\cdot}H^+{\cdot}2Br_3^-]_n$ particles, the energy of reaction was computed as the total electronic energy difference and free energy difference of the products and reactants, which are described in the Supporting Information with more details.

The calculations were performed using Gaussian 09D[66] with a focus on $[HV^{2+}{\cdot}2Br_3^-]_2$ cluster dimer formation because of the extreme computational cost of DFT calculations.

RESULTS AND DISCUSSION

Cyclic Voltammetry. Figure 1a shows a cyclic voltammogram (CV) measured in a 1 M H_2SO_4 solution containing 50 mM $HVBr_2$ for the electro-oxidation of Br^-. The current started to flow at 0.74 V and abruptly increased to 11.3 nA until a potential of 0.82 V, at which point the oxidation current suddenly decreased to ca. 0.79 nA. When the potential was more positively biased than 0.93 V, ill-defined voltammetric behavior was observed, with random fluctuation in current.

We observed that the onset potential, E_{onset}, for the electro-oxidation of Br^- measured in a 1 M H_2SO_4 solution is 0.07 V more negative than in 0.33 M Na_2SO_4; we will also discuss electrochemistry of Br^- and HV^{2+} in 0.33 M Na_2SO_4 later in this Article. In the previous Article, we reported that E_{onset} attributed to the electro-oxidation of Br^- becomes more negatively shifted (i.e., Br^- becomes more easily electro-oxidized) as the concentration of H^+ increases, which would mainly result from the facilitated charge transfers from diffusional Br^- to Pt due to the need of more positive potential for an adsorption of Br^- on a Pt as pH decreases.[19] Stucky et al. reported that Br_3^- from the electro-oxidation of Br^- interacts strongly with HV^{2+} and forms an ionic solid complex, $[HV^{2+}{\cdot}2Br_3^-](s)$.[15]

$$3Br^-(aq) \rightarrow Br_3^-(aq) + 2e^- \quad (1)$$

$$HV^{2+}(aq) + 2Br_3^-(aq) \rightarrow [HV^{2+}{\cdot}2Br_3^-](s) \quad (2)$$

After formation of an insulating $[HV^{2+}{\cdot}2Br_3^-]$ film, $[HV^{2+}{\cdot}2Br_3^-]_f$ forms on the Pt UME and leads to the observed dramatic current drop. Moreover, the abnormal current fluctuation at a more positive potential would be associated with irregular desorption and subsequent formation of $[HV^{2+}{\cdot}2Br_3^-]_f$. In the reverse scan of the CV, the small reductive peak is associated with the electroreduction of $[HV^{2+}{\cdot}2Br_3^-]_f$ on the

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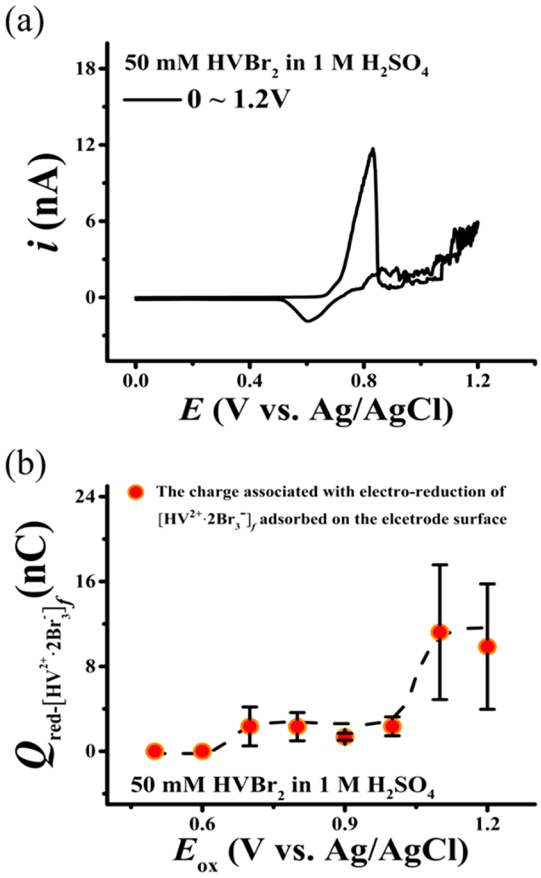

Figure 1. (a) CV for the electro-oxidation of Br^- to form $[HV^{2+}\cdot 2Br_3^-]$ on a Pt UME in 1 M H_2SO_4 aqueous solution containing 50 mM $HVBr_2$ and (b) $Q_{red}-[HV^{2+}\cdot 2Br_3^-]_f$ as a function of E_{ox}.

Pt UME. The peak separation of 0.69 V at the forward and reverse scan would result in the different reaction pathway in the electro-oxidation of Br^- from that in the electroreduction of Br_3^-, which are reported by Bard et al.[67,68]

$[HV^{2+}\cdot 2Br_3^-]$ would electrochemically form not only as a film but also as particles based on the estimation of the charge for the reduction peak at the reverse scan, Q_{red}, over that for the oxidation peak, Q_{ox}, associated with formation of $[HV^{2+}\cdot 2Br_3^-]$ in the forward CV (Figure 1a): $Q_{red}/Q_{ox} = 0.30$. The low value indicates that not all of the generated $[HV^{2+}\cdot 2Br_3^-]$ was electrochemically reduced within the potential window ($0 \leq E \leq 1.2$ V). The rest of $[HV^{2+}\cdot 2Br_3^-](s)$, which is not reduced in this potential region, would exist as particles, $[HV^{2+}\cdot 2Br_3^-]_p$; further evidence to strongly support the formation of $[HV^{2+}\cdot 2Br_3^-]_p$ is also well-described in the Supporting Information with Figures S1 and S2. In a previous article, our group reported the detailed electrochemical analyses for the formation of tetraalkylammonium tribromide (TBr_3) particles in the vicinity of Pt UME, which shows similar voltammetric behavior to the formation of $[HV^{2+}\cdot 2Br_3^-]_p$.[20] We will discuss the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$ through the impacts on modified Pt UME with $[HV^{+\bullet}\cdot Br^-]_f$ in acidic and neutral solutions.

Since the oxidative voltammetric behavior would be associated with the formation of both $[HV^{2+}\cdot 2Br_3^-]_f$ and $[HV^{2+}\cdot 2Br_3^-]_p$, it is difficult to solely extract electrochemical information on $[HV^{2+}\cdot 2Br_3^-]_f$ formation. To estimate more accurately the formation potential of $[HV^{2+}\cdot 2Br_3^-]_f$ on the Pt UME, we analyzed the charge associated with the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_f$ by integration of the reduction peak observed at the reverse scan as a function of the maximum oxidation potential, E_{ox}, in the CVs. The CV curves at various E_{ox} values are shown in Figure S3. The charge associated with the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_f$, $Q_{red}-[HV^{2+}\cdot 2Br_3^-]_f$, as shown in Figure 1b, started to increase from $E_{ox} = 0.70$ V and reached the first plateau with 2.46 nC. Then, $Q_{red}-[HV^{2+}\cdot 2Br_3^-]_f$ started to significantly increase again from $E_{ox} = 1.10$ V and reached the second plateau with 10.8 nC. From the report by Camacho et al.[69] and the reasonable consideration that the surface coverage of $[HV^{2+}\cdot 2Br_3^-]$ for a monolayer level would be the same as that of $[HV^{+\bullet}\cdot Br^-]$, we could estimate that the monolayer coverage of $[HV^{2+}\cdot 2Br_3^-]$ is 1.87×10^{-10} mol/cm^2, and the corresponding equivalent charge is 72 μC/cm^2, which is much smaller than the charge of the first plateau shown in Figure 1b. Therefore, it would be difficult to consider that the first plateau would be indicative of $[HV^{2+}\cdot 2Br_3^-]$ formation as a monolayer level. Rather than that, we believe that the observed first and second charge plateaus would be associated with the formation of $[HV^{2+}\cdot 2Br_3^-]_f$ strongly interacting with Pt at the first step and its subsequent bulk deposition.[70]

From the CV results in Figure 1a, the equilibrium constant of reaction 2 was roughly estimated. The total oxidation reaction (reactions 1 and 2) is as follows:

$$[HV^{2+}\cdot 2Br_3^-](s) + 4e^- \leftrightharpoons 6Br^- + HV^{2+} \tag{3}$$

The electrochemical half-cell potential of the reaction 1, $E_{(1)}$, is expressed as follows:

$$E_{(1)} = E_{(1)}^{0\prime} - \frac{0.05916}{2} \log \frac{C_{Br^-}{}^3}{C_{Br_3^-}} \tag{4}$$

where $E_{(1)}^0{}'$ is the formal potential of reaction 1, and both C_{Br^-} and $C_{Br_3^-}$ are the concentration of Br^- and Br_3^-, respectively. The equilibrium constant of reaction 2, $K_{eq(2)}$, is also expressed as follows:

$$K_{eq(2)} = \frac{1}{C_{HV^{2+}} C_{Br_3^-}{}^2} \tag{5}$$

where $C_{HV^{2+}}$ is the concentration of HV^{2+}. Therefore, the electrochemical half-cell potential for the net electrochemical reaction (reaction 3) through reactions 1 and 2, $E_{(3)}$, can be derived as follows, respectively:

$$E_{(3)} = E_{(1)}^{0\prime} - \frac{0.05916}{4} \log(C_{HV^{2+}}\cdot C_{Br^-}{}^6) - \frac{0.05916}{4} \log K_{eq(2)} \tag{6}$$

$E_{(3)}$ can also be expressed as follows:

$$E_{(3)} = E_{(3)}^{0\prime} - \frac{0.05916}{4} \log(C_{HV^{2+}}\cdot C_{Br^-}{}^6) \tag{7}$$

where $E_{(3)}^0{}'$ is the formal potential of reaction 3. From eqs 6 and 7, the following relation can be derived:

$$E_{(3)}^{0\prime} = E_{(1)}^{0\prime} - \frac{0.05916}{4} \log K_{eq(2)} \tag{8}$$

$E_{(3)}^0{}'$ could be roughly estimated as $(E_{pa} + E_{pc})/2$, where E_{pa} and E_{pc} are the oxidative and reductive peak potentials obtained from the CV results, respectively. $E_{(1)}^0{}'$ is ca. 0.828 V from the literature.[71] Therefore, $K_{eq(2)}$ was estimated to be 2.75×10^7 M^{-3}.

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The electrochemical formation of $[HV^{2+}\cdot 2Br_3^-]$ on Pt in both 1 M H_2SO_4 (pH ~ 0) and 0.33 M Na_2SO_4 (pH ~ 5.2) solutions was identified based on ATR-FTIR and Raman spectroscopy, as shown in Figures S4a,c and S4b,d. We confirmed that the chemical structure of electrochemically formed $[HV^{2+}\cdot 2Br_3^-]$ is identical in solutions containing either 1 M H_2SO_4 or 0.33 M Na_2SO_4; the detailed peak assignments are described in the Supporting Information. Also, we identified the presence of Br_3^- in electrochemically generated $[HV^{2+}\cdot 2Br_3^-]$ by XPS as well as shown in Figure S5.

The voltammetric behavior of formation of $[HV^{2+}\cdot 2Br_3^-]$ at the forward scan and the dissolution of $[HV^{2+}\cdot 2Br_3^-]_f$ at the reverse scan on a Pt UME in 0.33 M Na_2SO_4 aqueous solution without H_2SO_4 were ill-defined compared to those in 1 M H_2SO_4, as shown in Figure S6. From the CV, however, we can clearly observe that the peak associated with dissolution of $[HV^{2+}\cdot 2Br_3^-]_f$ was 0.16 V more negative than that in 1 M H_2SO_4 solution, indicating that electroreduction of $[HV^{2+}\cdot 2Br_3^-]_f$ to soluble HV^{2+} and $6Br^-$ becomes more electrochemically irreversible as pH increases, and the pH was ~5.2 in 0.33 M Na_2SO_4. The observed electrochemically irreversible behavior in the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_f$ to HV^{2+} and Br^- as a function of pH indicates the crucial role of H^+ for enhancement of reversibility in formation/dissolution of $[HV^{2+}\cdot 2Br_3^-]_f$. An initial elementary step of the overall electroreduction of $[HV^{2+}\cdot 2Br_3^-]$ would be considered a $1e^-$ transfer from Pt to $[HV^{2+}\cdot 2Br_3^-]$ to form $[HV^{2+}\cdot 2Br_3^-]^-$. To maintain charge neutrality in $[HV^{2+}\cdot 2Br_3^-]^-$, a cation in the aqueous phase should be diffused into the $[HV^{2+}\cdot 2Br_3^-]^-$ solid matrix. This proceeding cation transfer process coupled with the $1e^-$ transfer reaction would be the rate-determining step, and we believe that the H^+ transfer would be much faster than that of Na^+, resulting in increased reversibility. We will discuss the enhancement of reversibility for the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$ on a modified Pt UME with $[HV^{+\bullet}\cdot Br^-]_f$ depending on the pH of aqueous solutions affected by H^+ transfer in the next section.

Another type of dual-redox ionic solid can be formed by electroreduction of HV^{2+} in the presence of Br^-. First, the voltammetric behavior for the electroreduction of HV^{2+} was observed in a 0.33 M Na_2SO_4 aqueous solution, as shown in Figure 2a. In this condition, HV^{2+} started to be reduced at −0.48 V, and the corresponding reduction current abruptly increased to −4.83 nA and then decreased to −2.24 nA as the potential was negatively biased to −1.0 V.[72,73] At the reverse scan, the oxidation peak at −0.35 V was clearly observed. The reduction and oxidation peaks are attributed to the $[HV^{+\bullet}\cdot Br^-]$ ionic solid complexation as follows.[15]

$$HV^{2+}(aq) + e^- \leftrightharpoons HV^{+\bullet}(aq) \quad (9)$$

$$HV^{+\bullet}(aq) + Br^-(aq) \rightarrow [HV^{+\bullet}\cdot Br^-](s) \quad (10)$$

No further electroreduction of $HV^{+\bullet}$ species to HV^0 was observed in this condition, and this voltammetric behavior is distinctly different from that observed in a 1 M H_2SO_4 solution, which will be discussed later. The electrochemical formation of $[HV^{+\bullet}\cdot Br^-]$ on Pt in 0.33 M Na_2SO_4 solution (pH ~ 5.2) was confirmed by ATR-FTIR and Raman spectroscopy, as shown in Figure S7; the detailed peak assignments are described in the Supporting Information.

The ratio of Q_{ox} associated with the electro-oxidation of $[HV^{+\bullet}\cdot Br^-]$ to Q_{red} from its formation was only 0.20, indicating that $[HV^{+\bullet}\cdot Br^-]$ electrochemically forms as $[HV^{+\bullet}\cdot Br^-]_f$ and

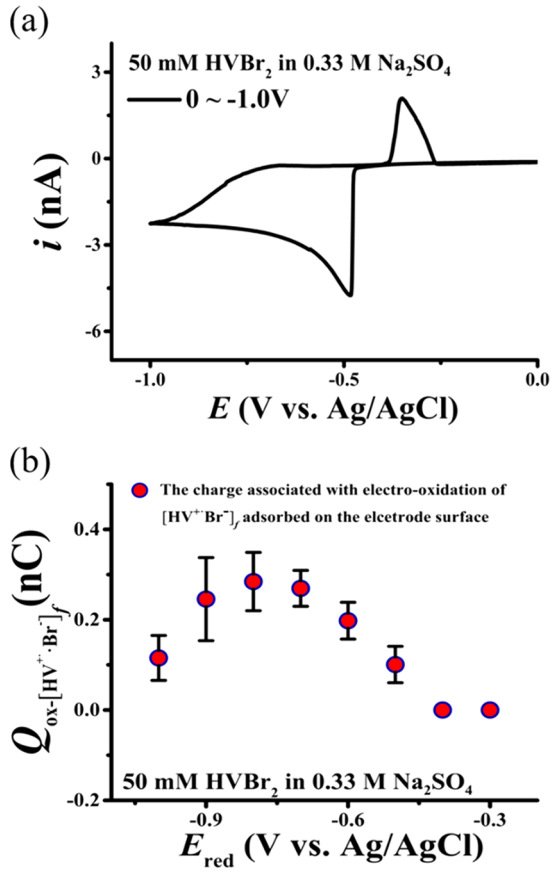

Figure 2. (a) CV for the electroreduction of HV^{2+} to form $[HV^{+\bullet}\cdot Br^-]$ in 0.33 M Na_2SO_4 and (b) corresponding $Q_{ox}-[HV^{+\bullet}\cdot Br^-]_f$ as a function of E_{red}.

$[HV^{+\bullet}\cdot Br^-]_p$. To extract electrochemical information on only $[HV^{+\bullet}\cdot Br^-]_f$, we estimated $Q_{ox}-[HV^{+\bullet}\cdot Br^-]_f$ as a function of E_{red}, as shown in Figure 2b. $Q_{ox}-[HV^{+\bullet}\cdot Br^-]_f$ started to increase from E_{red} = −0.45 V, reached the maximum at E_{red} = −0.8 V, and decreased to 0.12 nC as E_{red} became −1.0 V. From this plot, the maximum coverage of $[HV^{+\bullet}\cdot Br^-]_f$ was estimated as 3.75×10^{-9} mol/cm^2.

The equilibrium constant of reaction 10, $K_{eq(10)}$, was estimated from the CV in Figure 2a. The formation potential of the total reduction reaction from HV^{2+} to $[HV^{+\bullet}\cdot Br^-]$ (reaction 11) can be expressed as follows:

$$HV^{2+}(aq) + Br^-(aq) + e^- \leftrightharpoons [HV^{+\bullet}\cdot Br^-](s) \quad (11)$$

The electrochemical half-cell potential of reaction 9, $E_{(9)}$, is expressed as follows:

$$E_{(9)} = E_{(9)}^{0\prime} - 0.05916 \log \frac{C_{HV^{+\bullet}}}{C_{HV^{2+}}} \quad (12)$$

where $E_{(9)}^{0\prime}$ is the formal potential of reaction 9, and $C_{HV^{+\bullet}}$ is the concentration of $HV^{+\bullet}$. $K_{eq(10)}$ is also expressed as follows:

$$K_{eq(10)} = \frac{1}{C_{HV^{+\bullet}} C_{Br^-}} \quad (13)$$

Therefore, the electrochemical half-cell potential for the net electrochemical reaction (reaction 11) through reactions 9 and 10, $E_{(11)}$, can be derived as follows:

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$$E_{(11)} = E^{0\prime}_{(9)} + 0.05916 \log(C_{HV^{2+}}C_{Br^-}) + 0.05916 \log K_{eq(10)} \quad (14)$$

$E_{(11)}$ can be also expressed as follows:

$$E_{(11)} = E^{0\prime}_{(11)} + 0.05916 \log(C_{HV^{2+}}C_{Br^-}) \quad (15)$$

where $E^0_{(11)}{}'$ is the formal potential of reaction 11. From eqs 14 and 15, the following relation can be derived:

$$E^{0\prime}_{(11)} = E^{0\prime}_{(9)} + 0.05916 \log K_{eq(10)} \quad (16)$$

Here, $E^0_{(11)}{}'$ can be roughly estimated as $(E_{pa} + E_{pc})/2$ from the CV in Figure 2a. From eq 16, $K_{eq(10)}$ was calculated to be 57.3 M^{-2}.

The CV of the $HV^{2+}(aq)/[HV^{+\bullet}\cdot Br^-](s)$ redox reaction measured in a 1 M H_2SO_4 solution was different from that in the 0.33 M Na_2SO_4 solution with a pH of ~5.2, as shown in Figure 3a. At the forward scan of the CV, the reduction current

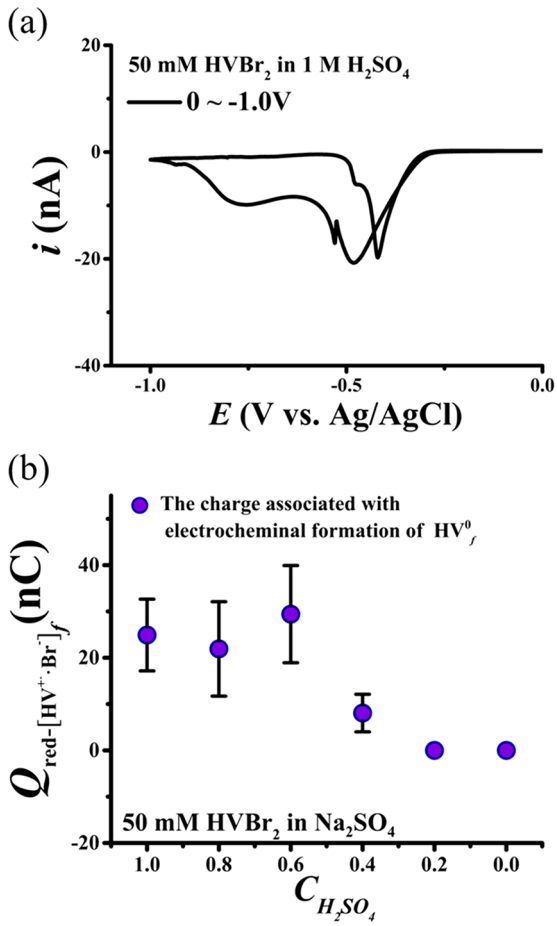

Figure 3. (a) CV for the electroreduction of HV^{2+} in a 1 M H_2SO_4 aqueous solution containing 50 mM $HVBr_2$ and (b) $Q_{red}-[HV^{+\bullet}\cdot Br^-]_f$ vs $C_{H_2SO_4}$ plot.

abruptly increased from −0.39 V and then started to decrease to −8.63 nA at −0.61 V. The hydrogen evolution reaction (HER) in this condition is completely inhibited. For a comparison, the CV was measured in a 1 M H_2SO_4 aqueous solution without $HVBr_2$, as shown in Figure S8. The reduction current associated with the HER became significant as the potential was negatively biased by more than −0.25 V. Therefore, the voltammetric characteristics result from formation of $[HV^{+\bullet}\cdot Br^-]$ on the Pt UME.

Interestingly, a broad reduction peak at −0.8 V was clearly observed, which was not observed in the CV obtained in the 0.33 M Na_2SO_4 solution. We found that this peak is strongly dependent on the concentration of H_2SO_4, $C_{H_2SO_4}$. Figure S10 shows enlarged CVs to observe the voltammetric response within the potential window between −0.65 and −0.95 V in solutions with various $C_{H_2SO_4}$ values. The peak started to be observed as $C_{H_2SO_4}$ approached 0.4 M and was maximized at $C_{H_2SO_4}$ = 0.8M, which indicates that protons mainly participate in the electrochemical reduction reaction. It is known that $HV^{+\bullet}$ is sequentially further reduced to HV;[74] therefore, we attribute this peak to sequential reduction of $[HV^{+\bullet}\cdot Br^-]$ to $[HV\cdot Br^-]$. Since the charge neutrality of $[HV\cdot Br^-]_f$ should be maintained, H^+ transfer to $[HV\cdot Br^-]_f$ would occur to form $[HV\cdot H^+\cdot Br^-]_f$ and eventually HV^0 as follows:

$$[HV^{+\bullet}\cdot Br^-]_f + e^- \leftrightharpoons [HV\cdot Br^-]_f \quad (17)$$

$$[HV\cdot Br^-]_f + H^+ \leftrightharpoons [HV\cdot H^+\cdot Br^-]_f \quad (18)$$

$$[HV\cdot H^+\cdot Br^-]_f \leftrightharpoons HV^0 \quad (19)$$

The reason for the absence of the reduction wave associated with further reduction of $[HV^{+\bullet}\cdot Br^-]_f$ in the 0.33 M Na_2SO_4 solution mainly results from the difficult Na^+ transfer into the $[HV^{+\bullet}\cdot Br^-]_f$ lattice due to the large size of Na^+.

Figure 3b shows the charge associated with the electroreduction of $[HV^{+\bullet}\cdot Br^-]_f$, $Q_{red}-[HV^{+\bullet}\cdot Br^-]_f$, as a function of $C_{H_2SO_4}$. $Q_{red}-[HV^{+\bullet}\cdot Br^-]_f$ started to increase at $C_{H_2SO_4}$ = 0.4 M and reached a plateau with $Q_{red}-[HV^{+\bullet}\cdot Br^-]_f$ = 29.4 nC at $C_{H_2SO_4}$ = 0.6 M. The maximum coverage of HV^0_f was estimated to be 3.88×10^{-7} mol/cm^2 from this plot, which is ca. 100 times higher than that of $[HV^{+\bullet}\cdot Br^-]_f$ formed in a 0.33 M Na_2SO_4 aqueous solution. Also, we roughly estimated the maximum flux of H^+ into $[HV^{+\bullet}\cdot Br^-]_f$ on the Pt UME as 2.22×10^{-8} mol/(cm^2 s) based on the voltammetric time of 17.5 s, which was measured from the potential range of the reduction wave attributed to the electroreduction of $[HV^{+\bullet}\cdot Br^-]_f$ to HV^0_f and the scan rate of the CV of 0.02 V/s.

At the reverse scan of the CV, no peak from the electrooxidation of HV^0_f was observed, whereas a sharp reductive peak was present. This abnormal voltammetric behavior could be explained as follows. As the potential is swept positively more than −0.37 V at the reverse scan, HV^0_f is oxidized, resulting in exposure of the Pt surface. Simultaneously, diffusive HV^{2+} is electrochemically reduced on the exposed Pt electrode to form $[HV^{+\bullet}\cdot Br^-]$. Since the oxidative current from HV^0_f is smaller than the reductive current from the electroreduction of HV^{2+} to form $[HV^{+\bullet}\cdot Br^-]$, only the reduction peak was observed in the CV at the reverse scan. This explanation is supported by the CVs measured at different $C_{H_2SO_4}$ values, as shown in Figure S11. The oxidation peak at the reverse scan was not observed from the CV measured in aqueous solutions containing more than 0.2 M H_2SO_4.

We tried to identify the formation of $[HV^{+\bullet}\cdot Br^-]$ on Pt in a 1 M H_2SO_4 aqueous solution through IR and Raman spectroscopy, similar to the case of $[HV^{2+}\cdot 2Br_3^-]$. However, the HER on a Pt macroelectrode was severe, and we could not detect $[HV^{+\bullet}\cdot Br^-]$ on the Pt surface. Also, the voltammetric behavior caused by the Pt macroelectrode no longer showed the unique feature of the Pt UME (see Figure S12). We believe that the different voltammetric feature for the electroreduction

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of HV^{2+} to form $[HV^{+\bullet}\cdot Br^-]$ in a 1 M H_2SO_4 solution on a Pt UME from that on a Pt macro-disk-electrode would be mainly because the formation of $[HV^{+\bullet}\cdot Br^-]$ would be much faster due to higher flux of HV^{2+} toward an edge of a Pt UME during its electroreduction (i.e., radial diffusion effect), and therefore, the HER would be effectively inhibited by $[HV^{+\bullet}\cdot Br^-]_f$ on a Pt UME.[75]

When we positively biased the potential for the electro-oxidation of Br^- at the forward scan and negatively swept the potential to reduce HV^{2+} at the reverse scan, we observed an interesting phenomenon from the CVs in both 1 M H_2SO_4 and 0.33 M Na_2SO_4 aqueous solutions, as shown in Figure 4a,b. In

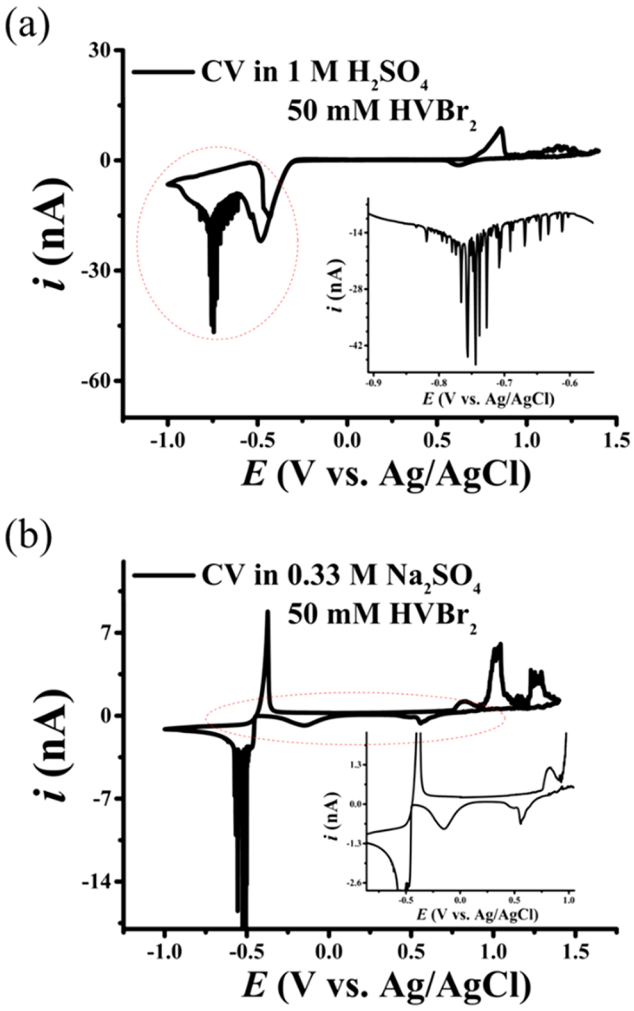

Figure 4. CVs measured in (a) 1 M H_2SO_4 and (b) 0.33 M Na_2SO_4 aqueous solutions containing 50 mM $HVBr_2$. The potential was initially swept from 0 to 1.4 V in the first segment. Then, the potential was negatively swept to −1.0 V in the second segment and finally returned to the initial potential. The enlarged CV is shown in the inset of the main graph.

the potential region, where HV^{2+} is electrochemically reduced, we observed numerous stochastic current spikes. In the next section, we present and discuss electrochemical analyses using chronoamperometry, which is a more suitable electrochemical method to analyze stochastic electrochemical events.[16−19]

Electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$ through Particle-Impact Analyses. Figure S13 shows a chronoamperogram (CA) of the electro-oxidation of Br^- in a 1 M H_2SO_4 aqueous solution containing 50 mM $HVBr_2$. In this $i-t$ curve, no stochastic current spikes were observed, and only current decay from Br^- oxidation to form $[HV^{2+}\cdot 2Br_3^-]$ was well-defined. After we applied 1.2 V for 60 s (henceforth, we designate this process as the pretreatment step), the potential was negatively stepped to various potentials, E_{red}, for another 60 s to electrochemically reduce HV^{2+} to $HV^{+\bullet}$ and then HV.

Figure 5 shows the corresponding CAs at different E_{red} values. Stochastic current spikes were observed as E_{red} became

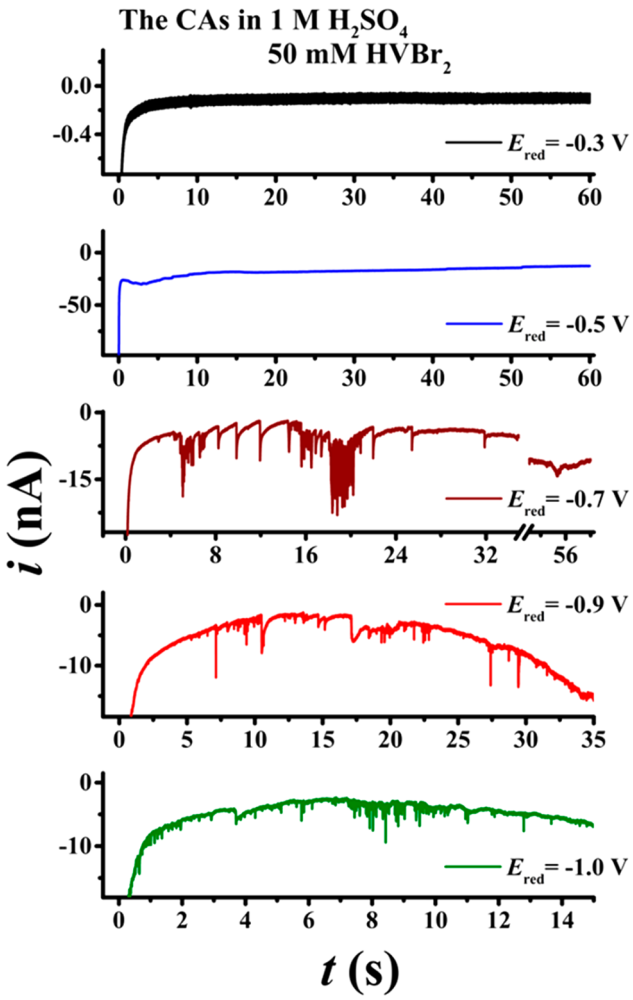

Figure 5. CAs at different E_{red} values in 1 M H_2SO_4 aqueous solutions containing 50 mM $HVBr_2$ after the pretreatment step for the generation of $[HV^{2+}\cdot 2Br_3^-]$.

greater than −0.6 V. These current spikes were not observed from the CA at the corresponding E_{red} without the pretreatment step for the electro-oxidation of Br^- to form $[HV^{2+}\cdot 2Br_3^-]$, as shown in Figure S14a. Based on our experimental results, we consider these current spikes to be due to the electrochemical reduction of electrochemically generated discrete $[HV^{2+}\cdot 2Br_3^-]_p$ through collision on the Pt UME.

We found that the predeposited $[HV^{2+}\cdot 2Br_3^-]_f$ on a Pt UME in the pretreatment step was completely dissolved within ca. 6 s at a constant $E_{red} \leq 0.2$ V in 50 mM $HVBr_2$ solutions with both 1 M H_2SO_4 and 0.33 M Na_2SO_4, as shown in Figure S15. Therefore, we posit that the reduction current at a constant potential in this time region (see Figure 5 and Figure S16) would be more complicated because the current is associated

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with simultaneous dissolution of $[HV^{2+}\cdot 2Br_3^-]_f$, electroreduction of HV^{2+}, and electroreduction of discrete $[HV^{2+}\cdot 2Br_3^-]_p$. However, electroreduction of both $[HV^{2+}\cdot 2Br_3^-]_f$ and HV^{2+} would only contribute to the background current in the CAs because of their continuum reduction characteristics, resulting in clear analysis of the current spikes even in this time region (≤6 s).

Figure 6 shows the average charges associated with the electroreduction of individual $[HV^{2+}\cdot 2Br_3^-]_p$ from CAs, $Q_{P,avg}$

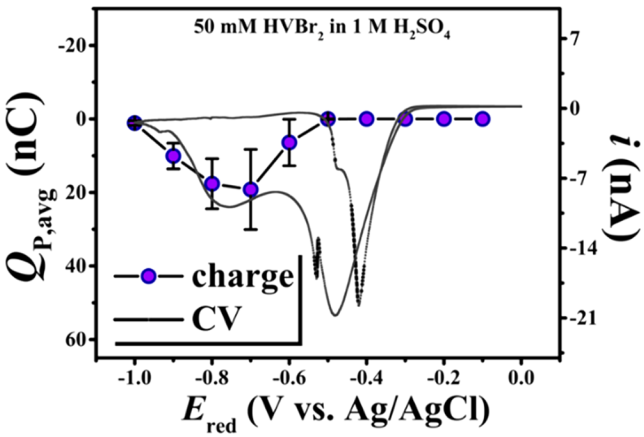

Figure 6. $Q_{p,avg}$ obtained at different E_{red} values overlaid on the CV in a 1 M H_2SO_4 aqueous solution containing 50 mM $HVBr_2$.

at different E_{red}, and overlaid plots for CVs measured in the same solution. $Q_{P,avg}$ starts to increase at E_{red} = −0.54 V and reached its maximum at E_{red} = −0.70 V. Then, $Q_{P,avg}$ decreased to 0 as E_{red} was negatively more polarized to −1.0 V. An $Q_{P,avg}$ of 0 in this plot means that we did not observe current spikes from the CA at the corresponding E_{red}. This different $Q_{P,avg}$ is attributed to the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$ at an applied potential during their collisions on the Pt UME, and the $Q_{P,avg}$ vs E_{red} plot could be considered the polarization curve for the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$. We estimated a maximum reduction charge of $[HV^{2+}\cdot 2Br_3^-]_p$ as ca. 19.3 nC in the 1 M H_2SO_4 aqueous solution. The polarization curve for the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$ is surprisingly well-matched with that of $[HV^{+\bullet}\cdot Br^-]_f$ to HV_f^0 via $[HV\cdot H^+\cdot Br^-]_f$, indicating that HV_f^0 has a crucial role in the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$.

We also investigated the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$ on a modified Pt UME with $[HV^{+\bullet}\cdot Br^-]_f$ in 0.33 M Na_2SO_4 solutions at different E_{red} values, as shown in Figure S16. Similar to the case in the 1 M H_2SO_4 aqueous solution, all the CAs were performed with the pretreatment step (Figure S17), and no stochastic current spikes were observed in the CAs associated with the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$ without the pretreatment step, as shown in Figure S14b.

When we observed the CAs with the pretreatment step, as in Figure S16, the dramatic difference compared to the CAs from the acidic solutions is that the number of stochastic current spikes within 60 s is sparse, which indicates that the probability for $[HV^{2+}\cdot 2Br_3^-]_p$ to be reduced on the modified Pt UME becomes significantly low as the pH of the solution increases to 5.

Figure 7 quantitatively supports the dependency of the stochastic electrochemical events of $[HV^{2+}\cdot 2Br_3^-]_p$ on solution pH. In an aqueous solution containing 1 M H_2SO_4, the

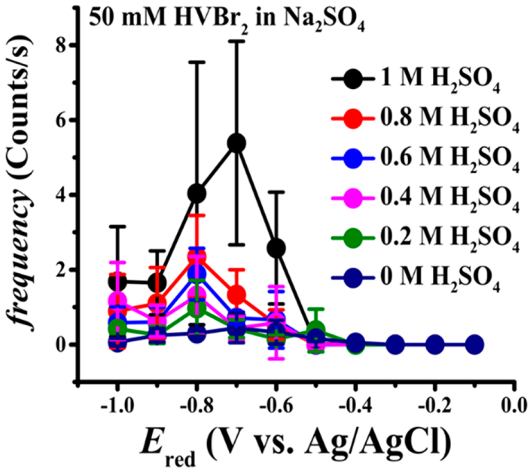

Figure 7. Frequency of stochastic current spikes from the CAs at different E_{red} values in solutions with various concentrations of H_2SO_4. I was maintained at 1 M by the addition of Na_2SO_4.

maximum collision frequency of $[HV^{2+}\cdot 2Br_3^-]_p$ is 5.38 (±2.72) Hz at E_{red} = −0.70 V. As $C_{H_2SO_4}$ reached 0.8 M, the maximum frequency significantly decreased to 2.34 (±1.11) Hz at E_{red} = −0.80 V. In a 0.33 M Na_2SO_4 solution without H_2SO_4, however, the maximum collision frequency of $[HV^{2+}\cdot 2Br_3^-]_p$ is only 0.44 (±0.40) Hz. The higher frequency of stochastic electrochemical events of $[HV^{2+}\cdot 2Br_3^-]_p$ in an acidic solution than that in a neutral solution indicates that the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$ on HV_f^0 would be kinetically faster than that on $[HV^{+\bullet}\cdot Br^-]_f$, and this kinetic difference infers a mechanistic change in the $[HV^{2+}\cdot 2Br_3^-]_p$ reduction process.

The evidence for the change of reaction pathway in the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$ on a modified Pt UME with $[HV^{+\bullet}\cdot Br^-]_f$ is also found from $Q_{p,avg}$ from the CAs measured in 0.33 M Na_2SO_4 solutions as a function of E_{red}, which is overlaid on the CV measured in the same solution in Figure 8.

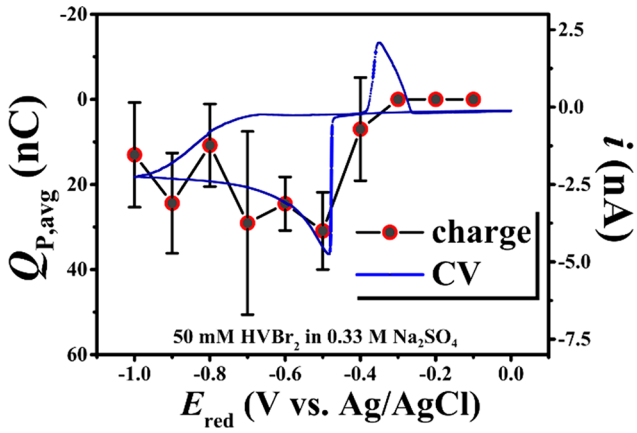

Figure 8. $Q_{p,avg}$ obtained at different E_{red} values overlaid on the CV obtained in a 0.33 M Na_2SO_4 aqueous solution containing 50 mM $HVBr_2$.

Although the error bars in each point are relatively large compared to those of $Q_{p,avg}$ measured in a 1 M H_2SO_4 solution, we confirmed that the trend of $Q_{p,avg}$ well follows the voltammetric behavior attributed to the formation of $[HV^{+\bullet}\cdot Br^-]_f$, indicating that $[HV^{2+}\cdot 2Br_3^-]_p$ is electrochemically

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reduced on the Pt UME with $[HV^{+\bullet}\cdot Br^-]_f$ in a 0.33 M Na_2SO_4 solution, while on Pt with HV_f^0 in 1 M H_2SO_4.

The change of the onset potentials (E_{onset}) from the polarization curve for $[HV^{2+}\cdot 2Br_3^-]_p$ reduction on a modified Pt UME in 50 mM $HVBr_2$ solutions containing different C_{H^+} values (Figure S18) more clearly shows the shift from $[HV^{+\bullet}\cdot Br^-]_f$ to $[HV\cdot H^+\cdot Br^-]_f$, which is galvanically coupled with $[HV^{2+}\cdot 2Br_3^-]_p$ as the pH decreases to 0.

Proposed Electrochemical Reduction Mechanism of $[HV^{2+}\cdot 2Br_3^-]_p$ in Neutral and Acidic Conditions. Based on the above analyses, we proposed the electrochemical mechanism of HV−Br ionic solid complexes as schematically described in Figure 9. In a neutral solution containing 0.33 M

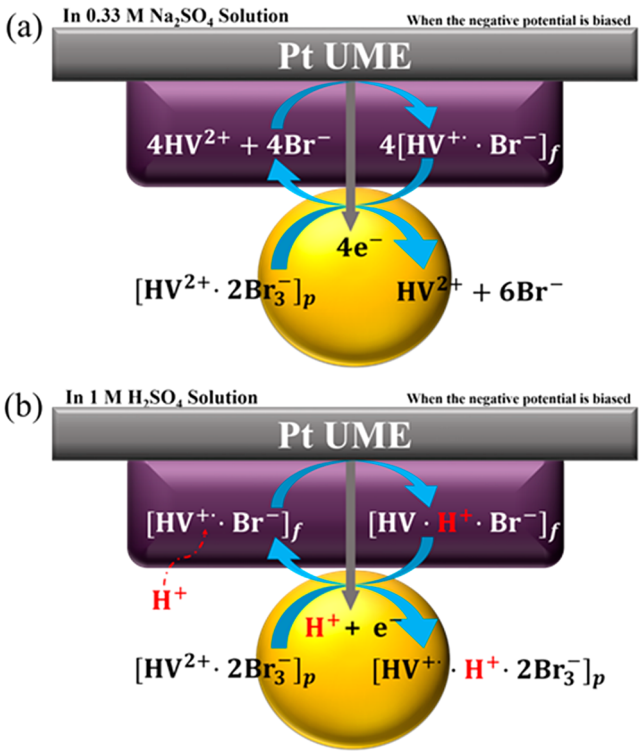

Figure 9. Schematic description of the proposed galvanic exchange reaction between two dual-redox ionic solids in (a) 0.33 M Na_2SO_4 and (b) 1 M H_2SO_4 aqueous solutions.

Na_2SO_4 with I = 1 M, $[HV^{2+}\cdot 2Br_3^-]_p$ is electrochemically formed on the Pt UME at a constantly applied 1.2 V and exists in the vicinity of Pt UME within our experimental time scale. As the potential was stepped to −0.8 V, $[HV^{+\bullet}\cdot Br^-]_f$ forms on the Pt UME. Since $[HV^{2+}\cdot 2Br_3^-]_p$ would stochastically collide on a modified Pt surface with $[HV^{+\bullet}\cdot Br^-]_f$, $[HV^{2+}\cdot 2Br_3^-]_p$ should interact with $[HV^{+\bullet}\cdot Br^-]_f$. The following galvanic exchange reaction, which is the total reaction of the two half redox reactions (reactions 3 and 11), is suggested in Figure 9a.

$$[HV^{2+}\cdot 2Br_3^-]_p + 4[HV^{+\bullet}\cdot Br^-]_f \rightarrow 5HV^{2+} + 10Br^- \quad (20)$$

Since we can estimate the dissolution potential of $[HV^{2+}\cdot 2Br_3^-]_p$ and formation of $[HV^{+\bullet}\cdot Br^-]_f$, it can be reasonably considered that the above galvanic exchange reaction is thermodynamically preferred (i.e., spontaneous). However, the suggested galvanic exchange reaction between $[HV^{2+}\cdot 2Br_3^-]_p$ and $[HV^{+\bullet}\cdot Br^-]_f$ involves $4e^-$ transfers overall, and cations should be diffused into each solid lattice to stabilize intermediates. In 0.33 M Na_2SO_4, a slow flux of Na^+ into the solid lattice is anticipated, resulting in a low frequency of stochastic current spikes.

When $C_{H^+(aq)}$ increases in an aqueous solution, the PIE event associated with the galvanic exchange reaction between two dual-redox ionic solids could occur via H^+ transfer, and the galvanic exchange reaction in a 1 M H_2SO_4 aqueous solution can be suggested as follows.

$$4[HV\cdot H^+\cdot Br^-]_f + [HV^{2+}\cdot 2Br_3^-]_p \leftrightharpoons 4[HV^{+\bullet}\cdot Br^-]_f + HV^{2+}(aq) + 6Br^-(aq) + 4H^+(aq) \quad (21)$$

In this case, electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$ would occur quickly because intermediates generated in each e^- transfer step could be stabilized by coupled H^+ transfer from $[HV\cdot H^+\cdot Br^-]_f$. However, H^+ is eventually liberated in the proposed reaction, which would be unfavored, especially in highly acidic conditions. The energy of reaction 21 computed at M062X/6-31G(d) in water solution amounts to be 382.27 kcal/mol (Table 1).

Table 1. Total Energies of the Reactants and Products of Reactions 21 and 22 and Their Energies of Reaction (FE in kcal/mol) Computed at M062X/6-31G(d) in Water

compound	reaction 21	reaction 22
$[HV\cdot H^+\cdot Br^-]$	−3618.326 00	−3618.326 00
$[HV^{2+}\cdot 2Br_3^-]_2$	−3295.524 10	−3295.524 10
HV·Br	−3618.250 05	−3618.250 05
$[HV\cdot H^+\cdot Br^-]_2$		−32 955.630 62
HV	−1046.178 00	
Br	−2571.919 03	
H^+	−0.215 18	−0.215 18
FE	382.27	−193.37

Therefore, we proposed the other galvanic exchange reaction via H^+ transfer, as schematically described in Figure 9b.

$$[HV\cdot H^+\cdot Br^-]_f + [HV^{2+}\cdot 2Br_3^-]_p \rightarrow [HV^{+\bullet}\cdot Br^-]_f + [HV^{+\bullet}\cdot H^+\cdot 2Br_3^-]_p \quad (22)$$

In this proposed reaction, $[HV^{2+}\cdot 2Br_3^-]_p$ is electrochemically reduced via $1e^-$−$1H^+$ transfer.

$$[HV^{2+}\cdot 2Br_3^-]_p + H^+ + e^- \leftrightharpoons [HV^{+\bullet}\cdot H^+\cdot 2Br_3^-]_p \quad (23)$$

It would be difficult to identify the reduction potential of reaction 23 from electrochemical measurements on a Pt UME because the current response from the electroreduction of $[HV^{2+}\cdot 2Br_3^-]_p$ to $[HV^{+\bullet}\cdot H^+\cdot 2Br_3^-]_p$ would be significantly perturbed by that associated with the formation of $[HV^{+\bullet}\cdot Br^-]_f$ and HV_f^0 via $[HV\cdot H^+\cdot Br^-]_f$. Therefore, we rationalize the spontaneity of reaction 22 through a computational approach, where the optimized geometries and the total electronic energies of $[HV^{2+}\cdot 2Br_3^-]_2$ and $[HV^+\cdot Br^-]$ and their related protonated species computed at B3LYP and M062X level are provided in the Supporting Information. The energy of reaction 22 was estimated to be nearly −193.77 kcal/mol at M062X/6-31G(d) in a water solvent (Table 1). In this calculation, the $[HV^{2+}\cdot 2Br_3^-]_p$ structure was considered a cluster dimer for simplicity. This finding indicates that the galvanic exchange reaction between $[HV\cdot H^+\cdot Br^-]_f$ and $[HV^{2+}\cdot 2Br_3^-]_p$ via $1e^-$−$1H^+$ transfer proceeds spontaneously. Also, the molecularity of reaction 22 is only 2, as opposed to the 5 of

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reaction 21. This difference indicates that reaction 22 would be kinetically more preferred, which results in a higher observed frequency of stochastic current spikes in an acidic solution than in a mild pH condition (0.33 M Na_2SO_4 solution).

Effect of Other Cations on the Redox Chemistry of [HV^{2+}·2Br_3^-] and [$HV^{+\bullet}$·Br^-]. Lastly, we noted that electrochemistry of [HV^{2+}·2Br_3^-] and [$HV^{+\bullet}$·Br^-] in aqueous solutions containing electrolytes with different cations (Li^+, Na^+, and K^+) are similar to each other, while their electrochemical behaviors are distinctively different from those in acidic solutions. Figure S19a,b shows the CVs measured in either a 0.33 M Li_2SO_4 aqueous solution or K_2SO_4 with the same concentration. The electroreduction of [HV^{2+}·2Br_3^-]$_f$ in both the electrolytes is electrochemically irreversible, which is similarly observed in a Na_2SO_4 solution. In addition, the CVs in both the Li_2SO_4 and K_2SO_4 solution (Figure S19c,d) do not show the peak attributed to the electroreduction of [$HV^{+\bullet}$·Br^-]$_f$ to [HV·C^+·Br^-]$_f$, which is analogous to that in Na_2SO_4; C^+ refers to the cation in each electrolyte.

Furthermore, the collision frequencies resulting from the electroreduction of [HV^{2+}·2Br_3^-]$_p$ on a modified Pt UME with [$HV^{+\bullet}$·Br^-]$_f$ via the stochastic collision of particles (Figure S20) show that even Li^+ would not involve the galvanic exchange reaction between [HV^{2+}·2Br_3^-]$_p$ and [$HV^{+\bullet}$·Br^-]$_f$.

These observations would support H^+ transfer in the electroreduction of both [HV^{2+}·2Br_3^-]$_p$ and [$HV^{+\bullet}$·Br^-]$_f$, and also the galvanic exchange reaction occurring between the two different dual RICs.

CONCLUSION

In summary, we present the electrochemistry for the formation of [HV^{2+}·2Br_3^-] and [$HV^{+\bullet}$·Br^-], their redox reaction, and the galvanic exchange reactions between them as a model system of dual-redox ionic solid complexes, which are critical for inhibition of the self-discharge reaction occurring in energy storage devices with redox electrolytes, such as Redox-EC.

For the formation of both [HV^{2+}·2Br_3^-] and [$HV^{+\bullet}$·Br^-], the charges associated with their formation are higher than those for their dissolution, which are deposited (or adsorbed) on a Pt UME. The observed unmatched charges demonstrate that both [HV^{2+}·2Br_3^-] and [$HV^{+\bullet}$·Br^-] would exist both as a film and as particles in the vicinity of a Pt UME.

Second, we found that the electrochemical reduction of both [HV^{2+}·2Br_3^-]$_f$ and [$HV^{+\bullet}$·Br^-]$_f$ is strongly influenced by $C_{H^+(aq)}$. For [HV^{2+}·2Br_3^-]$_f$, its reduction peak in the CV becomes more well-defined and reversible at pH = 0. This observation strongly indicates that H^+ involves the electroreduction of [HV^{2+}·2Br_3^-]$_f$. For [$HV^{+\bullet}$·Br^-]$_f$ as the other case, a new peak in the CV emerges as $C_{H^+(aq)}$ increases, which can be attributed to formation of HV_f^0 via H^+ transfer.

Third, stochastic current spikes are observed when [HV^{2+}·2Br_3^-]$_p$ collides on a modified Pt UME with either [$HV^{+\bullet}$·Br^-]$_f$ or HV_f^0 depending on $C_{H^+(aq)}$. As $C_{H^+(aq)}$ increases, the frequency of stochastic current spikes significantly increases, which indicates that the galvanic exchange reaction between the two dual-ionic solid complexes involves H^+ transfers. We propose the possible galvanic exchange reactions in aqueous solution containing either 0.33 M Na_2SO_4 or 1 M H_2SO_4. In a 0.33 M Na_2SO_4 solution, dissolution of the two dual-redox ionic complexes would occur. However, as $C_{H^+(aq)}$ increases to 1 M, the galvanic exchange reaction would be changed, and [HV^{2+}·2Br_3^-]$_p$ would be electrochemically reduced to [$HV^{+\bullet}$·H^+·2Br_3^-]$_p$ via the $1e^- - 1H^+$ transfer process in the proposed mechanism, which is rationalized based on the computational calculation of ΔG in the suggested galvanic exchange reaction. The suggested reaction via $1e^- - 1H^+$ would be kinetically more favored compared to the dissolution reaction with a $4e^-$ process, which explains our experimental results.

Lastly, the proton is revealed to be the cation among H^+, Li^+, Na^+, and K^+, which can involve the electroreduction of both [HV^{2+}·2Br_3^-]$_f$ and [$HV^{+\bullet}$·Br^-]$_f$, and also the galvanic exchange reaction between [HV^{2+}·2Br_3^-]$_p$ and [$HV^{+\bullet}$·Br^-]$_f$.

Our finding indicates that, even in the Redox-EC to use the dual redox-active ions, HV^{2+} and Br^- to form the two different types of dual RICs during the charging process, the self-discharge would occur if an average pore size of a separator in the Redox-EC would be larger than an average size of [HV^{2+}·2Br_3^-]$_p$ generated on a cathode, and this self-discharge would become more significant as the pH's of both a catholyte and an anolyte decrease.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.9b13985.

Experimental details, CVs, ATR-FTIR, Raman spectra, photographs, CAs, details of the DFT-optimized structures and calculation methods, and values associated with formation energy of HVBr (PDF)

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Notes

The authors declare no competing financial interest.

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